RESTRICTION GIVES A SHORT EXACT SEQUENCE OF GALOIS GROUPS IN A TOWER OF FINITE GALOIS EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The displayed sequence is the inflation-restriction starting point; the chapter uses it for the Galois correspondence in towers.

Editorial note. The label lemma-ses-galois names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-lift-maps (delivered card L48358: automorphisms restrict to normal subextensions and K-maps into a normal extension extend), lemma-multiplicativity-degrees (delivered card L48338), lemma-finite-Galois (a finite Galois group has order equal to the degree)

Source reference. Original reference 0BME: sources/stacks-fields/pinned-original-source.tex, source lines 2163–2163. Prerequisite pointer invoked at 3052; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09G9: sources/stacks-fields/pinned-original-source.tex, source lines 759–759. Prerequisite pointer invoked at 3059; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $L/M/K$ be a tower of fields. Assume L/K and M/K are finite Galois. Then we obtain a short exact sequence*

$$1 \to Gal(L/M) \to Gal(L/K) \to Gal(M/K) \to 1$$

of finite groups.

Proof. Namely, by Lemma 0BME we see that every element $\tau \in \mathrm{Gal}(L/K)$ induces an element $\tau|_M \in \mathrm{Gal}(M/K)$ which gives us the homomorphism on the right. The map on the left identifies the left group with the kernel of the right arrow. The sequence is exact because the sizes of the groups work out correctly by multiplicativity of degrees in towers of finite extensions (Lemma 09G9). One can also use Lemma 0BME directly to see that the map on the right is surjective. □

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